

**Time:** 5 minutes**Outcome:** Compare private and social incentives and identify the efficient quantity.**Difficulty:** ●●○ Intermediate

Externalities



THE CORE IDEA

An externality affects someone outside a transaction and is not reflected in its price. Marginal social cost (MSC) includes marginal private cost (MPC) and external cost. Marginal social benefit (MSB) includes marginal private benefit (MPB) and external benefit. Efficiency compares MSB with MSC; private choices compare MPB with MPC.



HOW TO RECOGNIZE IT

- Ask whether the spillover is a cost or benefit and whether it comes from production or consumption.
- Negative: $MSC > MPC$ or $MPB > MSB$, so the market quantity is too high.
- Positive: $MSB > MPB$ or $MPC > MSC$, so the market quantity is too low.
- Corrective taxes, subsidies, permits, regulation, or bargaining can internalize the spillover.

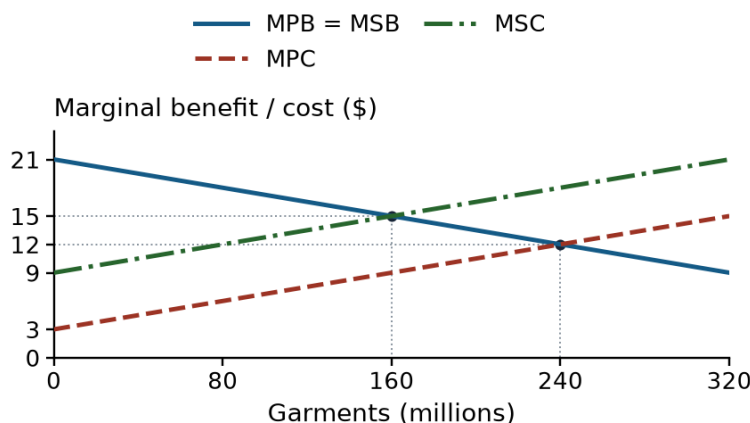


WATCH OUT

Do not label every harmful outcome an externality. A spillover must affect a third party and be missing from the market price.



WORKED EXAMPLE: NEGATIVE PRODUCTION EXTERNALITY



Fast-fashion production creates a \$6 external cost per garment, so MSC lies \$6 above MPC . The market produces 240 million garments where $MPB = MPC$ at \$12. Efficiency occurs at 160 million garments where $MPB = MSC$ at \$15, eliminating 80 million units of overproduction. A \$6 corrective tax aligns private and social marginal cost.



CHECK YOURSELF

If MSB lies above MPB , is the unregulated market quantity too high or too low?

**READY?**

Return to the game and master this concept.